
Part 3: Market Failures Practice Problem Solutions

Econ 502: Advanced Microeconomics

Problem 1: Adverse Selection in Insurance

Part (a): Certainty equivalent and maximum premium

Without insurance, expected utility for a type- i consumer is

$$EU_i = (1 - p_i)\sqrt{w} + p_i\sqrt{w - L} = (1 - p_i)(10) + p_i(8) = 10 - 2p_i$$

The **certainty equivalent** CE_i is the certain wealth that yields the same utility as the gamble:

$$u(CE_i) = EU_i \implies \sqrt{CE_i} = 10 - 2p_i \implies CE_i = (10 - 2p_i)^2$$

Plugging in:

$$CE_L = (9.6)^2 = 92.16, \quad CE_H = (8.8)^2 = 77.44$$

The consumer is indifferent between full insurance at premium $P_i^{\max}$ and going uninsured when $u(w - P_i^{\max}) = EU_i$, i.e., $w - P_i^{\max} = CE_i$:

$$\boxed{P_i^{\max} = w - CE_i}$$

$$P_L^{\max} = 100 - 92.16 = 7.84, \quad P_H^{\max} = 100 - 77.44 = 22.56$$

Verification: $P_H^{\max} = 22.56 > 7.84 = P_L^{\max}$. ✓

Note that each willingness-to-pay exceeds the corresponding actuarially fair premium $p_i L$ ($p_L L = 7.20$, $p_H L = 21.60$). The gap is the **risk premium** — the extra amount the risk-averse consumer is willing to pay above the expected loss to avoid the gamble.

Part (b): Pooled premium at $\lambda = 0.5$

$$P_{\text{pool}} = (0.5 \cdot 0.6 + 0.5 \cdot 0.2) \cdot 36 = 0.4 \cdot 36 = 14.40$$

- **Low-risk:** $P_L^{\max} = 7.84 < 14.40$. They will **not** buy.
- **High-risk:** $P_H^{\max} = 22.56 > 14.40$. They **will** buy.

If only high-risk types buy, the insurer's expected payout per policy is $p_H \cdot L = 21.60$, but it collects only 14.40 per policy. The insurer **loses \$7.20 per policy** and does not break even.

Part (c): Range of λ for unraveling

The pooled premium as a function of the high-risk share is

$$P_{\text{pool}}(\lambda) = (0.4\lambda + 0.2) \cdot 36 = 14.4\lambda + 7.2$$

Low-risk types drop out when $P_{\text{pool}}(\lambda) > P_L^{\text{max}}$:

$$14.4\lambda + 7.2 > 7.84 \implies \lambda > \frac{0.64}{14.4} = \frac{2}{45} \approx 0.044$$

Market unravels for $\lambda > 2/45 \approx 4.4\%$

Logic. At the pooled premium, low-risk types subsidize high-risk types. The maximum subsidy a low-risk type tolerates equals their risk premium ($P_L^{\text{max}} - p_L L = 0.64$). As soon as the pool contains enough high-risk types to push the pooled premium above P_L^{max} , low-risk consumers prefer to self-insure. Their exit raises the average risk in the pool, forcing the premium even higher — the classic adverse-selection death spiral. With only two types here, the dynamic stops once all low-risk are out and only high-risk remain (paying near $p_H L$).

Part (d): Compulsory insurance at $P_{\text{pool}} = 14.40$

Wealth under insurance is $w - P_{\text{pool}} = 85.60$ in both states, giving utility $\sqrt{85.60} \approx 9.252$.

Type	EU no insurance	u with mandate	Change
Low-risk	9.600	9.252	−0.348
High-risk	8.800	9.252	+0.452

- **Low-risk are worse off** (forced to subsidize the bad pool).
- **High-risk are better off** (get coverage at a price below their willingness to pay).

The insurer collects 14.40 per policy and pays out $0.5(0.6) + 0.5(0.2) = 0.4$ in probability $\times L = 36$ on average = 14.40, so it **breaks even**.

Why the policy can still be desirable. A utilitarian planner adds the changes: net gain = $-0.348 + 0.452 = +0.104$ utils per person — total welfare rises. More fundamentally, the mandate solves the adverse-selection failure: high-risk types receive coverage they could otherwise obtain only at much higher prices, while low-risk types lose only their small risk-aversion benefit. There are also equity arguments: people generally do not choose their risk type, and pooling spreads the cost of bad luck. This logic underlies, e.g., the individual mandate in the Affordable Care Act.

Problem 2: Pigouvian Tax and Cap-and-Trade**Part (a): Privately optimal output**

Each firm equates P to its private MC :

- Firm 1: $40 = 2q_1 \implies q_1^{\text{priv}} = 20$

- Firm 2: $40 = 4q_2 \Rightarrow q_2^{\text{priv}} = 10$

$$Q^{\text{priv}} = 30, \quad \text{damage} = 8 \cdot 30 = 240$$

Part (b): Socially efficient output

The planner adds the marginal external damage of 8 to each firm's MC :

- Firm 1: $40 = 2q_1 + 8 \Rightarrow q_1^* = 16$
- Firm 2: $40 = 4q_2 + 8 \Rightarrow q_2^* = 8$

$$Q^* = 24, \quad \text{damage} = 8 \cdot 24 = 192$$

Deadweight loss from unregulated production. For each firm, the social $MC + d$ exceeds P over the overproduced range:

$$DWL_1 = \int_{16}^{20} (2q + 8 - 40) dq = [q^2 - 32q]_{16}^{20} = (-240) - (-256) = 16$$

$$DWL_2 = \int_8^{10} (4q + 8 - 40) dq = [2q^2 - 32q]_8^{10} = (-120) - (-128) = 8$$

$$DWL = 16 + 8 = 24$$

Part (c): Pigouvian tax

With per-unit tax t , each firm now faces the maximization $\max_{q_i} Pq_i - C_i(q_i) - tq_i$, with FOC $P - t = MC_i$. Setting $t = 8$ (the marginal external damage):

- Firm 1: $32 = 2q_1 \Rightarrow q_1 = 16$ ✓
- Firm 2: $32 = 4q_2 \Rightarrow q_2 = 8$ ✓

$$t^* = 8 \text{ decentralizes the efficient outcome}$$

The tax forces each firm to internalize the full social cost of the externality. The price instrument applies to both firms uniformly because the marginal damage per unit is the same.

Part (d): Cap-and-trade

The cap is $\bar{Q} = Q^* = 24$ permits, with $\bar{e}_1 = \bar{e}_2 = 12$.

Firm i chooses q_i to maximize $Pq_i - C_i(q_i) - \tau(q_i - \bar{e}_i)$. The FOC gives $P = MC_i + \tau$, so each firm produces at the level where $MC_i = P - \tau$.

$$q_1 = \frac{40 - \tau}{2}, \quad q_2 = \frac{40 - \tau}{4}$$

The permit market clears when total demand equals the cap, $q_1 + q_2 = 24$:

$$\frac{40 - \tau}{2} + \frac{40 - \tau}{4} = \frac{3(40 - \tau)}{4} = 24 \implies 40 - \tau = 32 \implies \tau^* = 8$$

Plugging back: $q_1 = 16$, $q_2 = 8$ — exactly the efficient allocation.

Trades. Firm 1 needs 16 permits but starts with 12, so it **buys 4** from firm 2 (which holds 12 but uses only 8).

Comparison with the Pigouvian tax. The equilibrium permit price equals the Pigouvian tax rate ($\tau^* = t^* = 8$), and the equilibrium quantities are identical. The two instruments are dual: the tax fixes the price of pollution and lets the quantity adjust; cap-and-trade fixes the quantity and lets the market price adjust. Under the certainty assumed here, they produce the same outcome.

Problem 3: Coase Theorem

Part (a): Total surplus

	Doctor not SP	Doctor SP
Bakery noisy	260	300
Bakery quiet	280	250

The efficient combination is **bakery noisy + doctor soundproofed** (total surplus 300). Soundproofing eliminates a \$70 noise damage at a cost of only \$30, while letting the bakery keep its higher-profit noisy machinery.

Part (b): Doctor has the right to silence

Without bargaining. The bakery is forced to be quiet. The doctor compares 130 (no soundproofing) with 100 (soundproofing) and chooses **no soundproofing**. Outcome: (quiet, no SP), with payoffs $(\pi_B, \pi_D) = (150, 130)$ and total surplus 280.

With bargaining. The parties move to (noisy, SP), where the doctor agrees to soundproof and the bakery pays the doctor X for permission to be noisy. Joint surplus rises by 20.

- Bakery accepts if $200 - X \geq 150 \implies X \leq 50$
- Doctor accepts if $100 + X \geq 130 \implies X \geq 30$

$$30 \leq X \leq 50$$

Part (c): Bakery has the right to be noisy

Without bargaining. The bakery chooses to be noisy. The doctor compares 60 (no soundproofing) with 100 (soundproofing) and chooses to **soundproof**. Outcome: (noisy, SP), total surplus 300 — already efficient. **No bargaining is required.**

The doctor unilaterally adopts the efficient response because the cost of soundproofing (\$30) is less than the noise damage avoided (\$70).

Part (d): Comparison and the Coase theorem

In both regimes, the efficient outcome (noisy, SP) is reached with frictionless bargaining. **The assignment of property rights does not affect efficiency, only the distribution of payoffs:**

Regime	π_B	π_D
Doctor's right (bargained, $X = 30$)	170	130
Doctor's right (bargained, $X = 50$)	150	150
Bakery's right (no bargaining)	200	100

This is the **Coase theorem**: with well-defined property rights and zero transaction costs, private bargaining yields an efficient allocation regardless of the initial assignment of rights.

Why the assignment of rights still matters in practice:

- Real bargaining has **transaction costs** (lawyers, time, search) that may exceed the gains from trade.
- **Asymmetric information** about the parties' valuations can prevent agreement (each side may misrepresent reservation values).
- **Wealth and credit constraints**: if the loser cannot afford to pay the winner, the inefficient status quo persists.
- **Many parties** (multiple residents, multiple polluters) make Coasian bargaining infeasible due to free-riding and coordination costs.
- **Distributional / equity** considerations: even if efficient, society may prefer one assignment over another.

Problem 4: Public Good Provision

Part (a): Best response and Nash equilibrium

Person i chooses g_i to maximize

$$U_i = (Y_i - g_i)(g_i + g_{-i}) = (60 - g_i)(g_i + g_{-i})$$

The FOC is $-(g_i + g_{-i}) + (60 - g_i) = 0$, giving

$$g_i^*(g_{-i}) = 30 - \frac{g_{-i}}{2}$$

In a symmetric Nash equilibrium $g_1 = g_2 = g_3 = g$ and $g_{-i} = 2g$:

$$g = 30 - g \implies g_i^* = 15$$

$$G^* = 45, \quad x_i^* = 45$$

Part (b): Social optimum

Total welfare $\sum_i x_i G = XG$, with $X = \sum_i x_i$ and the resource constraint $X + G = 180$. Substituting,

$$\max_G G(180 - G) \implies G^{**} = 90$$

The equal allocation gives $X^{**} = 90$ and

$$x_i^{**} = 30, \quad G^{**} = 90$$

Part (c): Samuelson and the free-rider gap

For $U_i = x_i G$, $MRS_i = (\partial U_i / \partial G) / (\partial U_i / \partial x_i) = x_i / G$.

- **Nash:** $MRS_i = 45/45 = 1$ for each i . Each person privately equates MRS_i to the price of the public good, but $\sum_i MRS_i = 3$ — so collectively they massively under-provide.
- **Social optimum:** $MRS_i = 30/90 = 1/3$ for each i , and $\sum_i MRS_i = 1$. This is the **Samuelson condition** $\sum_i MRS_i = MRT$. ✓

$$\frac{G^*}{G^{**}} = \frac{45}{90} = \frac{1}{2}$$

Generalizing. With N symmetric contributors, the symmetric Nash satisfies $g_i^* = (60 - (N-1)g_i^*)/2$, giving $g_i^* = 60/(N+1)$ and $G^* = 60N/(N+1)$. The social optimum gives $G^{**} = NY/2 = 30N$. So

$$\frac{G^*}{G^{**}} = \frac{2}{N+1}$$

As $N \rightarrow \infty$, this ratio $\rightarrow 0$. The free-rider problem becomes overwhelming as the number of contributors grows.

Part (d): Pigouvian-style subsidy

With subsidy rate s , the consumer's budget is $x_i = 60 - (1-s)g_i$ and

$$U_i = (60 - (1-s)g_i)(g_i + g_{-i})$$

FOC: $-(1-s)(g_i + g_{-i}) + (60 - (1-s)g_i) = 0$, giving $60 = (1-s)(2g_i + g_{-i})$.

Symmetric: $60 = (1-s) \cdot 4g \implies g = 15/(1-s)$ and $G = 45/(1-s)$.

Setting $G = G^{**} = 90$:

$$\frac{45}{1-s} = 90 \implies 1-s = \frac{1}{2} \implies s^* = \frac{1}{2}$$

The government covers half the cost of each contribution, exactly internalizing the externality each contributor imposes on the other two. (More generally, for N contributors, $s^* = (N-1)/(N+1)$.)

Problem 5: Lindahl Prices

Part (a): Efficient public-good level

$\partial U_i / \partial x_i = 1$ and $\partial U_i / \partial G = \alpha_i / G$ with $\alpha_1 = 10$, $\alpha_2 = 6$. So

$$MRS_1 = \frac{10}{G}, \quad MRS_2 = \frac{6}{G}$$

The Samuelson condition $\sum_i MRS_i = MRT = 1$ gives

$$\frac{10}{G} + \frac{6}{G} = 1 \implies \boxed{G^{**} = 16}$$

Part (b): Lindahl prices

Consumer i takes τ_i as given and chooses G to maximize

$$U_i = (Y_i - \tau_i G) + \alpha_i \ln G$$

FOC: $-\tau_i + \alpha_i / G = 0 \implies G_i^d(\tau_i) = \alpha_i / \tau_i$.

For the Lindahl mechanism to deliver $G^{**} = 16$, we need $G_i^d(\tau_i^*) = 16$ for both consumers:

$$\tau_1^* = \frac{10}{16} = \frac{5}{8}, \quad \tau_2^* = \frac{6}{16} = \frac{3}{8}$$

Check: $\tau_1^* + \tau_2^* = 5/8 + 3/8 = 1 = MC$. ✓

$$\boxed{\tau_1^* = 5/8, \quad \tau_2^* = 3/8}$$

Part (c): Verification and intuition

At $\tau_1 = 5/8$: $G_1^d = 10 \div (5/8) = 16$. ✓ At $\tau_2 = 3/8$: $G_2^d = 6 \div (3/8) = 16$. ✓

Both consumers, individually, demand exactly G^{**} at the Lindahl prices. The mechanism works because each consumer is charged a personalized price equal to their own marginal willingness-to-pay at the efficient quantity. Setting individual demand equal to the efficient quantity ($\alpha_i / \tau_i = G^{**}$) is equivalent to requiring $\tau_i = MRS_i$, and summing gives back the Samuelson condition $\sum_i \tau_i = \sum_i MRS_i = MC$. So a personalized-price equilibrium reproduces efficient provision.

Part (d): Why it is hard in practice

Lindahl pricing requires the planner to know each consumer's MRS_i , which is **private information**. Each consumer has a strict incentive to **understate** how much they value G — claiming a low MRS lowers their personalized price while still benefiting from the public good (free-riding through misreporting).

This is the classic **demand-revelation problem** for public goods. Mechanisms like the Vickrey-Clarke-Groves (VCG) tax can elicit truthful reports as a dominant strategy, but only at the cost of running a budget surplus or deficit. There is no mechanism that simultaneously achieves efficiency, truth-telling in dominant strategies, and budget balance (a consequence of the Gibbard-Satterthwaite-style impossibility results for non-dictatorial mechanisms).